Measuring Coordination-Collapse Boundaries in Reward-Mixed Cooperative MARL

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Abstract

Cooperative multi-agent reinforcement learning (MARL) can collapse into degenerate *lazy-agent* policies when each agent's local reward overwhelms the team signal. We study this failure in reward-mixed cooperative MARL, where each learner is trained on $x_i(w) = (1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}$, and ask *where*, along the reward-centralization axis w, coordination gives way to collapse. We introduce a reusable boundary instrument: pre-specify a task-level collapse label, train across a grid of w, fit $P(\mathrm{collapse} \mid w)$, and report the critical centralization w_{crit} with bootstrap confidence intervals. Using this instrument, we test prospect-theoretic reward transforms applied after reward blending. *Loss aversion* shifts w_{crit} rightward: agents trained under loss-averse transforms require more team-level reward signal to coordinate. The direction replicates across four environment-algorithm combinations, the finite-horizon boundary slides left with additional training, and a mean-field account supports the sign; near a commonly used moderate loss-aversion setting ($\lambda \approx 3$), the shift is $+0.089$ net of curvature ($+0.188$ in isolation). On the primary supply-chain substrate, the instrument also exposes scoped secondary findings: value-function curvature lowers the boundary, a bounded affine law fits the loss-aversion sweep, and a single downstream loss-averse node can nearly set the boundary. These findings are substrate-specific unless replicated. The contribution is a measurement protocol for reporting where a cooperative learning system crosses into collapse, rather than only asserting that centralization helps.

Introduction

A team of learning agents can fail even when the task has a good cooperative solution: instead of learning complementary behavior, each agent settles into a locally safe, low-cooperation policy, and the team lands in a stable, useless equilibrium. We call this *coordination collapse*; in the MARL literature its endpoint is the *lazy-agent* policy (Liu et al. 2023; Shi and Peng 2023), and the underlying private-versus-collective conflict is the one studied in sequential social dilemmas (Leibo et al. 2017). We study this failure through a *reward-centralization* dial: within each reward-mixing sweep, agent i is trained on the blended scalar $x_i(w) = (1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}$, $w \in [0,1]$, so at $w{=}0$ the learner sees only its own local reward and at $w{=}1$ only the team reward. Turn w down and an agent's private performance drowns out its contribution; cooperating earns no learnable signal and the lazy policy wins. The field knows this direction (*more centralization helps* (Akella 2025; Liu et al. 2023)). What is missing is the location: lazy-agent and credit-assignment work mitigates the pathology without locating a boundary, and the closest phase-diagram studies place their critical points on *environmental* or opinion-dynamics axes rather than reward structure; none estimates a confidence-intervalled crossing in reward-centralization space for gradient-trained cooperative policies. Our central question is therefore twofold: does coordination collapse admit a *measurable* critical centralization threshold w_{crit}, and do prospect-theoretic reward transforms systematically shift it? The first half is a measurement problem (the paper's primary contribution), and the second is a question the instrument then makes answerable; we differentiate against the closest prior work below.

Pinning down this number for a given training setup matters because reward design is not neutral. w_{crit} is not universal: it is a measured property of an environment, learner, horizon, and collapse label. That is precisely what makes it useful: it tells a designer where their own system crosses into collapse. Our primary testbed is automated replenishment, where local inventory decisions and global service outcomes conflict; more broadly, any cooperative learning system that blends local and team rewards, or trains with preference-shaped objectives and human-motivated reward transforms (Ethayarajh et al. 2024; Lalmohammed et al. 2025), faces the same design question. We use prospect theory only as a controlled reward-transform family, not as a psychological claim: after local and team rewards are blended, the scalar is transformed with value-function curvature and loss aversion, the asymmetry in which a loss looms larger than an equal gain (Kahneman and Tversky 1979; Tversky and Kahneman 1992), while the learner, architecture, optimizer, and evaluation label are unchanged. The experimental question is then clean: does changing the scalar training signal move the collapse boundary?

We therefore build the ruler first, and treat it as the paper's central object. Our protocol is simple. Before fitting any boundary we fix a task-specific binary collapse label, a

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behavioral evaluation rule specified in advance (low/no ordering with poor profit in the supply chain, low landmark coverage in MPE, failed cooperative loading in LBF), then sweep w, fit $P(\text{collapse} \mid w)$, and bootstrap confidence intervals on w_{crit}. "Phase boundary" is used only in this operational sense, an empirical crossing in learning dynamics, not thermodynamic criticality. Two further design choices keep the readings interpretable. First, the prospect-theoretic value transform is applied after reward blending rather than inside the optimizer, so the architecture and optimization code are identical and matched runs differ only in the scalar training signal. Second, because a finite-horizon boundary can slide with training time, we re-measure w_{crit} at several horizons. The contributions are: (i) a confidence-intervalled boundary instrument for reward-mixed cooperative MARL; (ii) an application showing that loss aversion shifts the boundary rightward, agents trained under loss-averse transforms require more centralization to coordinate, with that direction replicating across four environment-algorithm combinations and the horizon slide generalizing; (iii) sign-level mechanism support under a checked supply-chain PPO premise; and (iv) scoped secondary findings the instrument exposes on the supply-chain substrate, where value-function curvature lowers the boundary, a bounded affine law fits the loss-aversion sweep, and a single downstream loss-averse node can nearly set the boundary. These findings are enabled by, but not required for, the main claim, and are reported as substrate-specific unless separately replicated.

Related Work

Coordination collapse and credit assignment. Cooperative MARL, studied since independent Q-learning on shared tasks (Tan 1993; Buşoniu, Babuška, and De Schutter 2008), inherits the lazy-agent and relative-overgeneralization failures in which agents abandon cooperation for a degenerate policy (Liu et al. 2023; Shi and Peng 2023; Tampuu et al. 2017). The standard remedy is centralized credit assignment under CTDE, a centralized critic or factorized value function (Lowe et al. 2017; Yu et al. 2022; Sunehag et al. 2018; Rashid et al. 2018; Foerster et al. 2018; Son et al. 2019); related work ties the failure to decentralized exploration (Weng and Lee 2026) or shows decentralized shaping cannot match centralized baselines (Akella 2025). This establishes *that* centralization helps and *how* to implement it; none locates the threshold w_{crit} or gives it a confidence interval, the gap we close, using w as the dial rather than a fixed factorization.

Phase transitions in multi-agent learning. Treating collective learning as a dynamical system with critical points has precedent: Yamaguchi (2025) builds a phase diagram for independent Q-learning over *environmental* axes, Capannoli et al. (2026) prove a metastable boundary in opinion dynamics under a fixed update rule, and the mean-field programme for multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019; Leonte et al. 2025) derives Fokker–Planck descriptions of networked Q-learning. We borrow the phase-diagram measurement style but differ in object: ours is a *reward-structure* threshold with confidence intervals that a prospect-theoretic reward transform moves, and we use the mean-field machinery only for a sign-level account.

Behavioral preferences in cooperative AI. A thin bridge connects human preference structure to collective learning dynamics: Hughes et al. (2018) show inequity aversion improves cooperation within the cooperative-AI agenda (Dafoe et al. 2020; Hammond et al. 2025; Barfuss et al. 2020). Risk-sensitive RL optimizes tail outcomes (Qiu et al. 2021; Shen et al. 2023; Prashanth 2014; Tamar, Glassner, and Mannor 2015), distinct from loss aversion, a reference-point asymmetry, and prospect theory has entered single-agent RL with known calibration concerns (Prashanth et al. 2016; Rabin 2000; Tversky and Kahneman 1992). In the multi-agent setting, Mengesha, Sun, and Roy (2025) prove loss aversion shifts a tipping point in evolutionary game theory over fixed strategies, and Lalmohammed et al. (2025) bring CPT into cooperative MARL but fold it into the update, entangling its components. We differ at the level of *apparatus*: rather than proposing another behavioral model, we build a measured boundary and use it as a common instrument; the prospect-theoretic reward-transform study is its first application, not the instrument's whole contribution.

Preliminaries and Method

We study *reward-mixed* cooperative MARL: settings in which N agents share an environment under CTDE (Lowe et al. 2017) and each learner's scalar training reward is a blend of a local term and a team term. The restriction is deliberate: it yields a controllable reward-design axis. For agent i, the blended scalar is $x_i(w) = (1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}$, $w \in [0,1]$, so w is not an optimizer hyperparameter but a structural property of the reward design. We do not rescale r_i^{local} or r^{global} separately at each w; both remain in the environment's native units, so a given w means the same local/team mixture in every condition rather than hiding a per-condition normalization. When w is small, an agent can protect its local reward while contributing little to the team; in tasks with local–global conflict this makes the low-cooperation policy easier to learn than the coordinated one (Liu et al. 2023; Shi and Peng 2023). Unless stated otherwise, each experiment varies one dial at a time: the w-sweep holds α, β, λ, learner, environment, horizon, and collapse label fixed, and each of those quantities varies only in its own named sweep.

The boundary estimate follows four steps: fix the environment, learner, horizon, and collapse label; train matched policies across a grid of w; label each run from pre-specified *evaluation* behavior, not training reward; and fit a monotone binary model $P(\text{collapse} \mid w)$, reporting the midpoint w_{crit} with bootstrap confidence intervals.

The reward transform is prospect-theoretic in value-function form. For the blended scalar $x = (1-w)r_i^{\text{local}} + wr^{\text{global}}$, with zero as the reference point for gains and losses, we use $v(x) = x^{\alpha}$ for $x \geq 0$ and $-\lambda(-x)^{\beta}$ for $x < 0$, where $\alpha, \beta \in (0,1]$ govern value-function curvature and $\lambda > 1$ is loss aversion (α=β=1, λ=1 is linear utility). We do not model probability weighting; the experimental

transform consists only of curvature and loss aversion. We apply it after blending,

$$r_i^{\mathrm{PT}} = v\big((1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}\big)\,, \qquad (1)$$

the decisive difference from Lalmohammed et al. (2025), who fold CPT *into* the actor–critic update, entangling curvature with loss aversion and confounding any shift with a change to the optimizer. The optimizer and architecture are identical across matched conditions; only the scalar training signal changes. A measured shift is therefore attributed to the transform channel under study, while recognizing that reward scale, variance, and advantage shape are part of that channel.

A *collapse label* is a pre-specified binary evaluation rule marking runs whose behavior lies inside a low-cooperation region. In the supply chain, collapse is $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$ ($\bar{S}$ mean base-stock, $\bar{\pi}$ mean profit), which corresponds to low/no ordering with poor profit. In MPE, collapse is landmark occupancy below $\theta = 0.144$, i.e. agents cluster rather than cover landmarks; in LBF, collapse is failed cooperative loading / low food collection under the pre-specified evaluation threshold. The supply-chain label, analysis code, and probes were committed before execution; the anonymized bundle freezes them for review, with the de-anonymized commit hash released on acceptance. Threshold robustness holds among estimable $\pm 20\%$ variants (max $\Delta w_{\mathrm{crit}} = 0.015$; Supplement).

For each condition we fit $P(\text{collapse} \mid w) = \sigma(b_0 + b_1 w)$ and report $w_{\mathrm{crit}} = -b_0/b_1$, the w at which fitted collapse probability is 0.5. We bootstrap the 95% CI; steep high-λ cells use Firth penalized regression, whose w_{crit} agrees with the bootstrap estimate to within 0.019 at every λ, giving concordant estimation procedures on the same crossing. The transition width is the fitted 10–90 width, $\Delta w = (\mathrm{logit}(0.9) - \mathrm{logit}(0.1))/|b_1|$, reported alongside location. We use "phase boundary" in this empirical crossing sense. The decomposition compares three conditions with identical environment and optimizer: the **linear baseline** ($\alpha{=}\beta{=}1, \lambda{=}1$; no curvature, no loss aversion), the **curvature baseline** ($\alpha{=}\beta{=}0.88, \lambda{=}1$), and **loss-aversion conditions** ($\alpha{=}\beta{=}0.88, \lambda > 1$). The primary substrate is `NetworkManagement-v1` (Hubbs et al. 2020), a six-node multi-echelon inventory network (Oroojlooyjadid et al. 2022; Gijsbrechts et al. 2022; Leluc et al. 2023); transfer environments are MPE Cooperative Navigation (Lowe et al. 2017) and Level-Based Foraging (Christianos, Schäfer, and Albrecht 2020). We choose the primary supply-chain substrate because it exposes a directed-flow coordination failure and supports the minority-topology analysis; the boundary instrument itself only requires a reward-mixing dial and a pre-specified collapse label, and the *direction* is confirmed on MPE and LBF to rule out a supply-chain artifact. Agents run PPO (Schulman et al. 2017) via Stable-Baselines3 (Raffin et al. 2021) and PettingZoo/SuperSuit (Terry et al. 2021; Terry, Black, and Hari 2020). The 1,524 primary supply-chain runs cover structural boundaries, the null reproduction, decomposition, the loss-aversion sweep, basin control, and horizon checks; headline λ sweeps use 20 seeds/(w, λ) cell. Full setup is in the Supplement.

The Boundary and Its Reward-Transform Shift

A sharp boundary exists at fixed training budget. At 300k steps on the supply-chain PPO substrate, collapse has a clear boundary in reward-centralization space: the fitted midpoint is $w_{\mathrm{crit}} = 0.666$ $[0.635, 0.694]$. The transition is sharp rather than a gradual decay: fitted collapse probability rises from 10% to 90% over $\Delta w = 0.199$, and a Firth logistic-slope test rejects a flat relationship between w and collapse ($p = 3.2 \times 10^{-7}$; Figure 1a). We use 300k as the primary horizon because the risk-neutral boundary is interior and sharply estimable there; horizon dependence is measured separately below.

The boundary is finite-horizon and metastable. The boundary slides left with the training horizon: across four matched risk-neutral horizons, $w_{\mathrm{crit}} = 0.822, 0.666, 0.622, 0.573$ at $150\mathrm{k}, 300\mathrm{k}, 450\mathrm{k}, 900\mathrm{k}$ steps, and the slide is *log-linear in* T: $w_{\mathrm{crit}} = 2.44 - 0.138 \ln T$ (RMSE 0.027; all four CIs of width ≤ 0.073, the last three ≤ 0.054; Figure 1b). This is consistent with metastable escape (a Kramers-type first-passage picture, developed below): each doubling of the budget buys a fixed ≈ 0.10 leftward shift, so longer training rescues some runs that collapse at shorter horizons. An illustrative extrapolation to $2\mathrm{M}$ steps gives $w_{\mathrm{crit}} \approx 0.44$, but the claim is limited to the measured horizons; the deterministic floor $w_{\mathrm{crit}}^0 = c/(c+g)$ derived below is treated as lying beneath the measured range. Horizon dependence does not make w_{crit} arbitrary: at any fixed budget it is well-defined, and the slide quantifies the rate at which extra training substitutes for centralization.

The horizon slide is not a supply-chain artifact. Under centralized-critic MAPPO the same slide recurs in both transfer environments: on MPE w_{crit} falls $0.82 \to 0.68$ as the horizon grows $2\mathrm{M} \to 8\mathrm{M}$, and on Level-Based Foraging $0.33 \to 0.20$ over $1\mathrm{M} \to 4\mathrm{M}$, both with non-overlapping endpoint intervals. A basin-control experiment checks a simple confound, that runs collapse because their initial policies start closer to the lazy basin: within each fixed w, correlations between initial and converged base-stock are only small to moderate (-0.11 to -0.38) and collapsed versus surviving runs are not separated by their starting base-stock, so initialization is an unlikely driver of the measured boundary (Supplement).

Loss aversion raises the boundary; curvature lowers it. Three measured baselines anchor the comparison, all with identical environment, PPO learner, collapse label, and 300k horizon: the structural null of the main sweep ($\alpha{=}\beta{=}1, \lambda{=}1$; $w_{\mathrm{crit}} = 0.666$), the same linear condition independently reproduced at 0.685 for the decomposition (other null-like numbers are labeled offsets, not separate baselines), and the curvature baseline ($\alpha{=}\beta{=}0.88, \lambda{=}1$). We fix $\alpha{=}\beta{=}0.88$ because it is the conventional prospect-theory curvature estimate (Tversky and Kahneman 1992), not a value tuned to the task; the loss-aversion sweep then varies only $\lambda \in \{1, \ldots, 7\}$. The boundary rises monotonically up to estimation noise,

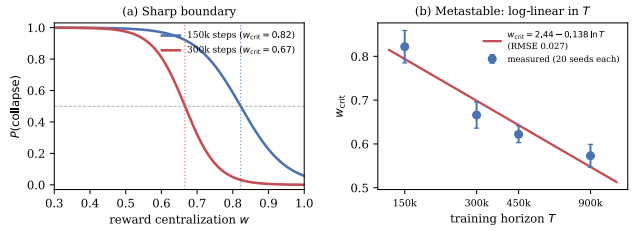


Figure 1: Boundary location and horizon slide. (a) At fixed training budget, collapse changes sharply with reward centralization w; curves show fitted $P(\text{collapse} \mid w)$ at 150k and 300k steps. (b) Increasing the training horizon shifts the boundary left, log-linearly in T (RMSE 0.027; 20 seeds per horizon): more training rescues some runs that collapse at shorter horizons.

$\{0.586, 0.715, 0.774, 0.806, 0.825, 0.835, 0.852\}$, then saturates near 0.85 with clear headroom to the w=1 ceiling, so the plateau is a reward-transform saturation, not a ceiling artifact (Figure 2, Table 1); the endpoints are non-overlapping by 0.221. Curvature alone acts in the *opposite* direction, *lowering* w_{crit} to 0.586 (-0.100 $[-0.172, -0.033]$ vs. the linear baseline). This is why we decompose the transform: reported only as a single "prospect-theoretic" treatment, the two effects would partially cancel, since on this substrate curvature makes coordination easier while loss aversion makes it harder.

Two effect sizes answer two questions. The sweep gives boundary locations; effect sizes subtract baselines, and the reported size depends on the comparison. The *deployment shift* compares a transformed reward with the linear baseline and is the total change a designer would see when replacing the linear reward with the full transform: at λ=3, $w_{\text{crit}} = 0.774$ is $+0.089$ over 0.685; at λ=7, $+0.166$. The *isolated loss-aversion shift* compares the same conditions with the curvature baseline (0.586) and is the cleaner estimate of the λ effect, because curvature is present in every sweep condition: $+0.188$ and $+0.265$ respectively (reconciled by the curvature offset, $-0.100 + 0.265 \approx 0.166$). We highlight λ=3 as a moderate, empirically motivated setting and λ=7 as the sweep's strongest value; the full series appears in Figure 2 and the Supplement. Throughout, "the loss-aversion effect" means the increment over the curvature baseline. The full sweep is tracked by a one-parameter, theory-bounded affine law (derived below) to RMSE 0.014, and a finer $w \in \{0.80, 0.90\}$ grid tightens the steep $\lambda \geq 5$ cells to CI widths ≈ 0.03.

Mechanism: Loss Aversion Lowers Escape SNR

The measurement above is the paper's contribution; this section adds a scoped mean-field account of the *sign* of the loss-aversion shift for the primary supply-chain PPO setting, under a local sign premise checked on that substrate. It uses continuous base-stock actions, the reward-mixing dial w, and the observed local–global incentive conflict. It does

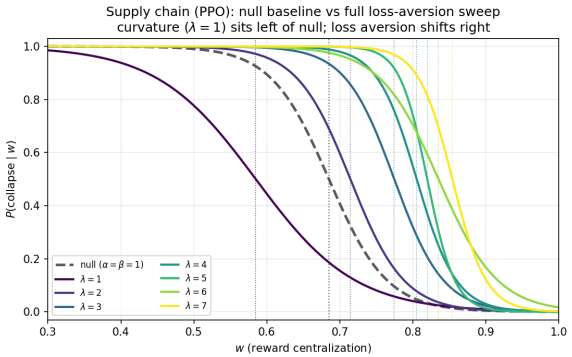


Figure 2: Loss aversion raises the collapse boundary (supply chain, PPO). Curves are fitted $P(\text{collapse} \mid w)$; dotted lines mark w_{crit}. Curvature alone (α=β=0.88, λ=1) shifts the boundary *left* of the linear baseline; increasing λ then shifts it *right*, saturating near $w_{\text{crit}} \approx 0.85$.

Table 1: Decomposing reward-transform shifts (supply chain, PPO, 300k). "vs. null" is the deployment shift against the reproduced linear baseline; "vs. curv." is the isolated loss-aversion shift against the curvature baseline. ✓ = difference CI excludes zero.

Condition	α, β	λ	w_{crit}	vs. null	vs. curv.
Linear null (rep.)	1, 1	1	0.685	–	–
Curvature	0.88, 0.88	1	0.586	$-0.100^{\checkmark}$	0
Loss av.	0.88, 0.88	3	0.774	$+0.089^{\checkmark}$	$+0.188$
Loss av.	0.88, 0.88	7	0.852	$+0.167^{\checkmark}$	$+0.266$

not predict the numerical value of w_{crit} in closed form. Following mean-field treatments of multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019; Leonte et al. 2025), the Supplement writes the Gaussian policy mean as a diffusion with drift F and update-variance Σ. Near the boundary, the relevant object is escape signal-to-noise, not drift alone:

$$\mathcal{S}(w, \lambda) = \frac{F_{\text{esc}}(w, \lambda)}{\sqrt{\Sigma_{\text{esc}}(w, \lambda)}}, \tag{2}$$

where the escape interval runs from the lazy basin to the coordinating basin. A finite-horizon boundary is the crossing $\mathcal{S}(w, \lambda) \approx \kappa_T$. In the Kramers first-passage picture (Kramers 1940), $\tau_{L \to C}$ is the time to escape the lazy basin into coordination, so collapse at horizon T means $\tau_{L \to C} > T$. Increasing w shortens that escape time; increasing loss aversion lengthens it through the transformed loss-domain signal.

The same account is consistent with an exploration-noise dissociation, not uniform movement everywhere. Injected entropy raises w_{crit} on the supply chain and LBF, where injected exploration is a binding noise term, but stays flat on MPE, where dense reward and intrinsic exploration dominate (Figure 3). This supports the SNR account while keeping its magnitude calibrated: two pre-committed probes re-

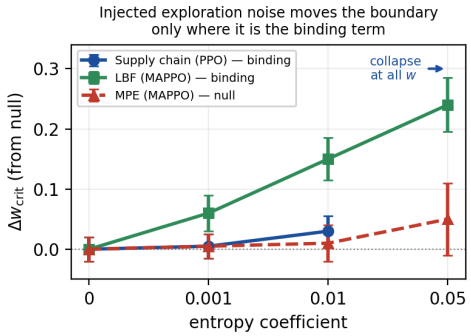


Figure 3: Entropy-ablation dissociation. Injected exploration noise raises w_{crit} on the supply chain (PPO) and LBF (MAPPO), where it is the binding noise term, but leaves the MPE boundary flat (overlapping CIs), where dense-reward intrinsic exploration dominates.

produced the boundary but did not localize it from drift or basin-flow statistics (Supplement).

The sign result is conditional on a local premise checked on the supply-chain substrate: increasing centralization raises escape SNR, while increasing loss aversion lowers escape SNR on the lazy-to-coordination interval. The loss-aversion effect may operate through drift, diffusion, or both; we therefore state the premise at the SNR level rather than identifying it with drift alone.

Proposition 1 (Loss aversion raises the empirical boundary)**.** *Fix a training horizon T and assume, near the boundary: (i) $\tau_{L\to C}(w,\lambda) = h(\mathcal{S}(w,\lambda))$ with $h, \mathcal{S}$ continuously differentiable and $h' < 0$; (ii) $\partial_w \mathcal{S} > 0$ and $\partial_\lambda \mathcal{S} < 0$; (iii) for each λ a unique interior $w^{\mathrm{Emp}}_{\mathrm{crit}}(\lambda)$ solves $\tau_{L\to C}(w,\lambda) = T$. Then $\mathrm{d}w^{\mathrm{Emp}}_{\mathrm{crit}}/\mathrm{d}\lambda > 0$.*

Proof. By (i)–(ii), $\partial_w \tau_{L\to C} = h'(\mathcal{S})\,\partial_w \mathcal{S} < 0$ and $\partial_\lambda \tau_{L\to C} = h'(\mathcal{S})\,\partial_\lambda \mathcal{S} > 0$. By (iii) the boundary solves $\tau_{L\to C}(w^{\mathrm{Emp}}_{\mathrm{crit}}(\lambda),\lambda) = T$ with $\partial_w \tau_{L\to C} < 0 \neq 0$, so the implicit function theorem gives a differentiable $w^{\mathrm{Emp}}_{\mathrm{crit}}(\lambda)$ with $\mathrm{d}w^{\mathrm{Emp}}_{\mathrm{crit}}/\mathrm{d}\lambda = -\partial_\lambda \tau_{L\to C}/\partial_w \tau_{L\to C} > 0$. □

The proposition fixes only the direction; the magnitude depends on unobserved diffusion scale and basin geometry.

The curvature result is scoped more narrowly. On the primary supply-chain PPO substrate, value-function curvature raises the escape SNR relative to the linear transform and lowers w_{crit}, while loss aversion lowers escape SNR and raises w_{crit} (Table 1); the transfer experiments below show the loss-aversion direction generalizes while the curvature sign does not.

A theory-bounded law, tested out of sample. The supply-chain PPO sweep is additionally summarized by a bounded affine law after anchoring the λ=1 curvature baseline. Only the loss-domain share of the marginal escape incentive scales with λ, giving

$$w_{\mathrm{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda)+\gamma}, \qquad \phi(\lambda) = 1 + \rho(\lambda - 1), \tag{3}$$

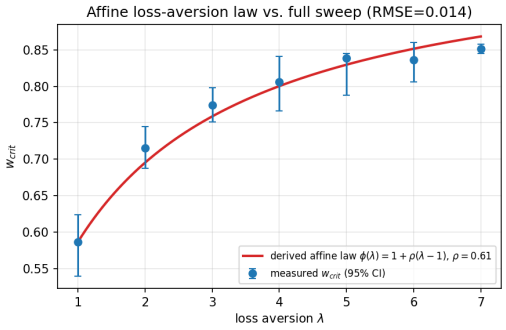


Figure 4: The bounded affine law (3) (line, one additional free parameter $\rho = 0.61$ after anchoring γ at λ=1) against measured $w_{\mathrm{crit}}(\lambda)$ (points, bootstrap 95% CIs). Held-out RMSE on $\lambda \in \{2,4,5,6\}$ is 0.016; all-seven RMSE is 0.014.

Table 2: Claim ledger for the mechanism. Each row separates modeled signs, measured evidence, and fitted quantities.

Modeled/sign-level	supply-chain PPO: increasing w raises escape SNR; increasing λ lowers escape SNR; loss aversion raises w_{crit}
Measured generally	sharp boundary; horizon slide; loss-aversion direction across SC, MPE, LBF
Measured only on SC	curvature lowers w_{crit}; affine law; directed-flow minority cascade
Calibrated/fitted	absolute w_{crit}; diffusion scale; γ; ρ after baseline anchoring
Future work	closed-form numerical w_{crit}; universal curvature sign; affine-law transfer

with $\rho \in [0,1]$. The baseline fixes γ; ρ is one additional calibrated parameter set from $\lambda \in \{3,7\}$, leaving $\lambda \in \{2,4,5,6\}$ held out: $w^{\mathrm{pred}}_{\mathrm{crit}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\}$ versus $w^{\mathrm{meas}}_{\mathrm{crit}} = \{0.586, 0.715, 0.774, 0.806, 0.825, 0.835, 0.852\}$. The held-out RMSE is 0.016; the all-seven RMSE is 0.014 (Figure 4). Leave-one-out fits keep ρ in $[0.59, 0.64]$ (mean 0.61, s.d. 0.02), a stability check rather than the original calibration split. The sign transfers; the affine form does not: the law is a single-substrate summary that fits the supply-chain PPO sweep and does not transfer to MPE or LBF (Table 4), so it should not be read as a universal MARL law. Table 2 states the evidence bucket for each mechanism claim.

The Instrument Travels; So Does the Direction

The first thing that must generalize is the *instrument*: the boundary protocol is only useful if it can be measured beyond the supply chain. It can: we estimate a CI'd w_{crit} in two further environments and under a second algorithm, and ask whether the loss-aversion direction travels; the mechanism

commits in advance to a sign through the SNR premise.

A transfer boundary is meaningful only if the learner can coordinate at high centralization, and independent PPO fails this prerequisite on the shared-reward transfer tasks: on MPE it collapses at *every* w, including maximal centralization (a coordination floor), and the same floor appears on LBF (Supplement), so no interior crossing exists to estimate. Centralized-critic MAPPO (Yu et al. 2022) recovers coordination and exposes a measurable boundary; we therefore use MAPPO for MPE and LBF. The design is thus not fully crossed, and to separate the learner change from the environment change we also run MAPPO on the supply-chain substrate with the same grid and label. On *MPE* (Cooperative Navigation) the team objective is landmark coverage; a herding wrapper adds a local reward for staying near other agents, creating the required local-versus-team conflict. Its scale is chosen so the no-transform baseline boundary is interior to the measured w-grid, and a run is labeled collapsed when landmark occupancy falls below the prespecified threshold $\theta = 0.144$.

The loss-aversion direction transfers in every measured setting (Figure 5, Table 4). On the *supply chain under MAPPO* (only the learner changes), w_{crit} rises $0.796 \rightarrow 0.889$, monotone from λ=2, converging with PPO by $\lambda \geq 4$; on *MPE* the loss-aversion excess is $+0.158$ at λ=7; on *LBF* w_{crit} rises $0.26 \rightarrow 0.83$ by λ=4. The secondary structure does not transfer: on MPE the curvature-only shift is indistinguishable from zero ($+0.042$, CI spans zero), and on LBF curvature moves the boundary *right* rather than left while the supply-chain affine law misfits the sweep (RMSE 0.15). The portable claim is the CI'd boundary and the loss-aversion direction; absolute levels, curvature sign, and affine form are substrate-specific.

Scope condition: action representation. On the supply-chain base-stock control, the signal-to-noise crossing depends on a continuous scalar action representation (Table 3). Value-based learners (QMIX, IQL) on a discretized K=11 base-stock shell give a flat, w-independent collapse rate, but a single-variable control isolates the cause: PPO on the *same* discrete shell also goes flat, so it is coarse scalar discretization, not the value-based paradigm, that erases the boundary. We interpret this as coarse bins snapping the policy to fixed attractor levels and suppressing near-boundary exploration; it is a scope condition of the supply-chain mechanism, not a claim that discreteness per se removes the boundary. MPE is natively discrete in action labels but remains a continuous spatial coordination task and retains a boundary. Since value-based learners show no measurable boundary here, whether the loss-aversion direction holds under them is not well-posed, and we claim nothing about it.

A Single Downstream Node Can Nearly Set the Boundary

The main sweep is homogeneous: every trained node uses the same λ. We now test a heterogeneous setting, measured only on the supply-chain directed-flow substrate: one node is made loss-averse while the other trained nodes remain risk-neutral, holding the environment, PPO configuration, collapse label, w-grid, and boundary estimator identical to the main sweep. We ask two questions: can one biased node move the boundary by itself, and does its position determine the size of the shift (Figure 6)? Position could matter because the six-node network is a branching DAG: demand enters at the retail edge, two downstream trained nodes feed demand most directly, and upstream suppliers feed one or both downstream paths, so a loss-averse node should matter more when its output reaches demand through a short, concentrated path. MPE and LBF do not have this directed resource-flow topology, so the cascade is a falsifiable directed-flow prediction rather than a tested generalization.

Table 3: Action-representation ablation. The control (last row) changes only the action space while holding the learner at PPO, isolating coarse discretization as what erases the boundary.

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes
QMIX (centralized value)	discrete K=11	no
IQL (independent value)	discrete K=11	no
PPO (control)	discrete K=11	**no**

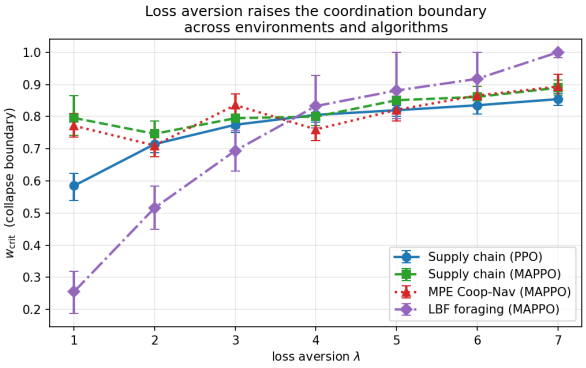


Figure 5: w_{crit} versus λ across four settings: supply chain under PPO and MAPPO, MPE under MAPPO, LBF under MAPPO (bootstrap 95% CIs). w_{crit} rises with λ in every setting; the *direction* generalizes, while levels (most at λ=1), curvature sign, and exact form are substrate-specific. LBF $\lambda \geq 5$ points extrapolate past the w=1 ceiling (outside the measured grid) and are labeled exploratory.

A single node, dose for dose. Sweeping the most downstream node's $\lambda \in \{1, \ldots, 7\}$ (others risk-neutral) traces a strictly monotone dose-response: w_{crit} rises $0.586 \rightarrow 0.856$, converging on the *fully* loss-averse value (0.852) by λ=7 (Figure 6b). One node, biased hard enough, nearly matches the entire network's midpoint requirement; the homogeneous result is the saturation of a continuous law, not a special case.

Position: proximity to demand, not chain index. Placing one λ=7 node at each position, the effect is strongly position-dependent, but the governing variable is how di-

Table 4: Generalization of the loss-aversion *direction*. Endpoints are reported as λ=1 $\rightarrow$ 7 where both are interior; LBF uses the last interior endpoint. Detail in the Supplement.

Setting (algo.)	Endpoints / shift	Affine law?
Supply (PPO)	$0.586 \rightarrow 0.852$ ($+0.266$ vs curv.)	yes
Supply (MAPPO)	$0.796 \rightarrow 0.889$ ($+0.093$)	–
MPE (MAPPO)	$0.78 \rightarrow 0.89$ ($+0.158$ excess)	no
LBF (MAPPO)	$0.26 \rightarrow 0.83$ by λ=4	no (0.15)

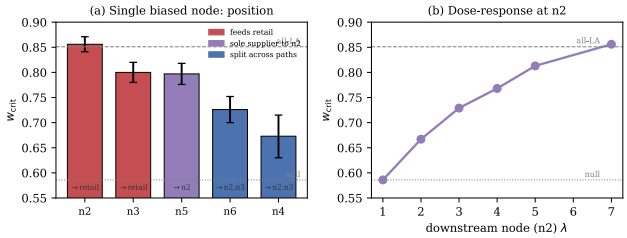


Figure 6: A single loss-averse node can dominate the supply-chain boundary. (a) One λ=7 node by position: the largest shifts come from nodes whose output reaches demand most directly, nodes feeding retail (red) dominate, the sole supplier to the dominant downstream node (purple) nearly matches them, and split-path suppliers (blue) are weaker. (b) Dose-response: increasing the downstream node's λ moves w_{crit} monotonically up to the homogeneous all-loss-averse value (0.852, dashed).

rectly a node's output reaches demand, not a chain index (Figure 6a). The two nodes feeding retail directly dominate (0.856, 0.800); the upstream supplier feeding *only* the dominant downstream node is nearly as strong (0.797), while those *splitting* output across both are weaker (0.726, 0.673). The network is a branching DAG, not a chain, and w_{crit} tracks a qualitative *path concentration* pattern: a biased node matters more the more directly and exclusively its product flows to demand (cf. the EGT tipping-point result of Mengesha, Sun, and Roy 2025, here in deep-MARL with a topological signature).

Discussion

That value-function curvature and loss aversion exert opposite effects on the primary substrate is, to our knowledge, novel; prior work applies prospect theory as a unified model without decomposing it (Lalmohammed et al. 2025; Ghaemi et al. 2024). The operational reading is concrete: a system of λ=7 reward-transformed agents has a 50% collapse midpoint at $w = 0.852$ against $w = 0.685$ for risk-neutral agents, a $+0.167$ midpoint margin that scales with λ; settings with effective λ near 2–3 would incur a smaller but positive shift under this substrate. Our phase-diagram framing sits within a recognized line of multiagent phase-transition results (Yamaguchi 2025; Capannoli et al. 2026) and the broader cooperative-AI project (Dafoe et al. 2020; Hammond et al. 2025; Barfuss et al. 2020); the distinctive contribution is a measured reward-centralization boundary with a supporting signal-to-noise account.

Operational interpretation. The boundary is a design quantity: a team can estimate its own w_{crit}, budget for the loss-aversion margin rather than discover collapse in production, and (where w is capped) treat an effective λ induced by the reward transform or training objective as a second lever to measure rather than assume (Ethayarajh et al. 2024; Rabin 2000). The minority result adds a targeting rule: audit the biased node whose output most directly reaches demand, not the average agent. The same topology information can be used defensively, to monitor or harden the most influential node, or adversarially, to identify where an induced bias would most degrade coordination; we report both openly so the effect stays auditable.

Scope and Future Work. Three boundaries each point to a concrete next step. *(i) From one substrate to many.* The quantitative core (the $w_{\mathrm{crit}}(\lambda)$ law and the cascade) is single-substrate by construction: the direction, boundary, and metastable slide replicate across environments and a second algorithm, but the magnitudes belong to one simulator and the affine law does not transfer (MPE, LBF). Replicating the cascade on a second directed-flow domain would make it a tested generalization. *(ii) From calibrated to predictive.* The mechanism fixes sign and form but not magnitude; localizing w_{crit} from the learning dynamics resisted two pre-committed probes, both of which track *where the policy sits*, not *where the boundary is* (Supplement). A localizing observable is the clearest open problem. *(iii) From bracket to calibration.* Estimating the effective λ of preference-aligned agents from behavior would tie the reward-transform axis to the training pipeline. None undercuts the central, replicated claim.

Conclusion

We introduced a confidence-intervalled boundary instrument for reward-mixed cooperative MARL: pre-specify a collapse label, sweep w, fit $P(\mathrm{collapse} \mid w)$, and report w_{crit} with error bars. Using it, we showed, with sign-level mechanism support, that loss-aversion reward transforms shift the boundary rightward across four environment-algorithm combinations: agents require more centralization to coordinate. The direction and the horizon slide generalize; the numerical thresholds, curvature sign, affine law, and minority cascade remain substrate-specific unless replicated. The protocol is mechanism-agnostic once a control dial and collapse label are specified; prospect-theoretic reward transformation is its first application.

Ethical Statement

This work is entirely simulated, with no human subjects or personal data. Reported thresholds are substrate-specific and must be re-measured before operational use; the minority-node result is dual-use (hardening versus targeted disruption) and reported explicitly for auditability.

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